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Direct current offset calibration device on the receiving path and direct current offset calibration method thereof

Abstract

A direct current (DC) offset calibration device and a method are used to calibrate a DC offset of a unit. The DC offset calibration device includes a signal generation unit, a to-be-calibrated unit, a measurement unit, and a compensation unit. The DC offset calibration method includes: using the to-be-calibrated unit to output a clipped signal resulting from a signal saturation effect; receiving a receiving signal correlated to the clipped signal, the receiving signal including an even harmonic resulting from the clipped signal; measuring a magnitude of the even harmonic to obtain a DC offset adjustment value accordingly; and adjusting the to-be-calibrated unit according to the DC offset adjustment value to calibrate the DC offset.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to electronic circuits, and in particular, to a DC offset calibration device on a receiving path and a DC offset calibration method thereof.

BACKGROUND

(2) In an electronic circuit, a signal may have a DC offset due to factors such as an impedance value error, thereby causing signal distortion. The circuit may not be able to maintain a good performance. In related art, a DC offset measurement circuit can be arranged in the signal transmission path of the communication circuit, so as to measure and detect the DC offset of the transmission signal, and accordingly calibrate or compensate for the DC offset. However, this may increase circuit complexity and circuit area. In addition, in some circuit designs, the DC offset

measurement circuit may not be necessary in the signal transmission path, and in this condition, although the DC offset may be adjusted manually, it is often time consuming and the DC offset may not be effectively eliminated as desired.

SUMMARY

(3) According to an embodiment of the invention, a direct current (DC) offset calibration device includes a signal generation unit, a to-be-calibrated unit, a measurement unit, and a compensation unit. The signal generation unit may provide a first signal. The to-be-calibrated unit is coupled to the signal generation unit, and configured to receive the first signal and output a second signal accordingly. The second signal is a clipped signal. The measurement unit is coupled to the to-be-calibrated unit and configured to receive a receiving signal correlated to the second signal. The receiving signal may comprise an even harmonic resulted from the clipped signal, and the measurement unit is configured to measure a magnitude of the even harmonic. The compensation unit is coupled to the measurement unit and the to-be-calibrated unit. The compensation unit is used to compute and obtain a DC offset adjustment value according to the magnitude of the even harmonic, and to adjust the to-be-calibrated unit according to the DC offset adjustment value.

(4) According to another embodiment of the invention, a DC offset calibration method is provided and the method may be used to calibrate a DC offset of a to-be-calibrated unit. The to-be-calibrated unit may output a second signal, and the second signal is clipped due to a signal saturation effect of the to-be-calibrated unit. The second signal may be clipped asymmetrically further due to the DC offset of the to-be-calibrated unit. A receiving signal correlated to the second signal may be received, and the receiving signal includes an even harmonic resulting from the asymmetrical clipping. A magnitude of the even harmonic may be measured. A DC offset adjustment value may be computed according to the magnitude of the even harmonic. Then, the to-be-calibrated unit may be adjusted according to the DC offset adjustment value to calibrate the DC offset.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram of a communication circuit according to an embodiment of the invention.

(2) FIG. **2** is a flowchart of a DC offset calibration method of the DC offset calibration device in FIG. **1**.

(3) FIG. **3** shows a waveform symmetrically clipped according to an embodiment of the present invention.

(4) FIG. **4** shows the frequency components of the waveform in FIG. **3**.

(5) FIG. **5** shows a waveform asymmetrically clipped according to an embodiment of the present invention.

(6) FIG. **6** shows the frequency components of the waveform in FIG. **5**.

(7) FIG. **7** shows the DC offset response of even harmonics at a first interval.

(8) FIG. **8** shows the DC offset response of even harmonics at a second interval.

DETAILED DESCRIPTION

(9) Below, exemplary embodiments will be described in detail with reference to accompanying drawings so as to be easily realized by a person having ordinary knowledge in the art. The inventive concept may be embodied in various forms without being limited to the exemplary embodiments set forth herein. Descriptions of well-known parts are omitted for clarity, and like reference numerals refer to like elements throughout.

(10) The terms “module” and “unit” as used herein generally represent software, firmware, hardware, or a combination thereof.

(11) FIG. **1** is a block diagram of a communication circuit **1** according to an embodiment of the

invention. The communication circuit **1** includes a direct current (DC) offset calibration device **10**, a transmitter **15**, a transmitting antenna **16**, a receiving antenna **17** and a receiver **18**. In some embodiments, the transmitting antenna **16** and/or the receiving antenna **17** may include conventional radio frequency antennas. For example, the antennas may include, but are not limited to, shortwave antennas, ultrashort wave antennas and microwave antennas depending on different operating frequency bands. The antennas may include, but are not limited to, omnidirectional antennas and directional antennas depending on different directivity. The antennas may also include, but are not limited to, linear antennas and planar antennas according to different shapes.

(12) In some embodiments, the DC offset calibration device **10** may include a signal generation unit **11**, a to-be-calibrated unit **12**, a measurement unit **13** and a compensation unit **14**. The signal generation unit **11**, the to-be-calibrated unit **12**, a transmitter **15** and a transmitting antenna **16** may be coupled in sequence to form part of a signal transmission path. The receiving antenna **17**, the receiver **18**, the measurement unit **13** and the compensating unit **14** may be coupled in sequence to form part of a signal receiving path. In other words, the signal generation unit **11** and the to-be-calibrated unit **12** may be arranged in the signal transmission path, and the measurement unit **13** and the compensation unit **14** may be arranged in the signal receiving path. In some embodiments, the measurement unit **13** and the compensation unit **14** may be integrated together.

(13) In the above embodiments, the signal generation unit **11** may provide a first signal Sa. The to-be-calibrated unit **12** may be coupled to the signal generation unit **11** and receive the first signal Sa. The to-be-calibrated unit **12** may output the second signal Sc according to the first signal Sa, and the second signal Sc is clipped. That is, the second signal Sc may be a clipped signal. Further, the clipped signal may be a symmetrical clipped signal or an asymmetrically clipped signal as shown in FIGS. **3** and **5** respectively, and description will be explained in the following paragraphs. The measurement unit **13** may be coupled to the to-be-calibrated unit **12**, and may receive a receiving signal Sr correlated to the second signal Sc. The receiving signal Sr includes even harmonics resulted from the clipped signal. The measurement unit **13** may measure a magnitude Mag of an even harmonic. The compensation unit **14** may be coupled to the measurement unit **13**, and may compute the DC offset adjustment value OFST according to the magnitude Mag of the even harmonic. The compensation unit **14** may be further coupled to the to-be-calibrated unit **12**, such that the to-be-calibrated unit **12** may receive the DC offset adjustment value OFST and may be adjusted accordingly, so as to reduce or eliminate the DC offset of the to-be-calibrated unit **12**.

(14) In some embodiments, the signal generation unit **11** may include a signal generator **110** and a signal amplifier **112**. The to-be-calibrated unit **12** may include a DC offset calibration circuit **120** and a to-be-calibrated circuit **122**. The signal generator **110**, the signal amplifier **112**, the DC offset calibration circuit **120**, the to-be-calibrated circuit **122** and the transmitter **15** may be coupled in sequence.

(15) In the above embodiments, the signal generator **110** may generate a reference signal Sref. In some embodiments, the signal generator **110** may include a sine wave generator, and the reference signal Sref may include a sine wave. The signal amplifier **112** may receive the reference signal Sref, and amplify the reference signal Sref based on a predetermined gain, so as to output the first signal Sa. In some embodiments, the first signal Sa may have a first amplitude. The signal amplifier **112** may include a variable gain amplifier. The gain of the signal amplifier **112** may be set differently for various to-be-calibrated units. For example, the gain of the signal amplifier **112** may be set to amplify the reference signal Sref generated from the signal generator **110**, such that the amplified signal, i.e., the first signal Sa, may have a first amplitude which exceeds the linear operating range of the to-be-calibrated unit **12**. In other words, the to-be-calibrated unit **12** is saturated. As a result, clipping occurs in the output signal (i.e., the second signal Sc) of the to-be-calibrated unit **12**. In this embodiment, for different to-be-calibrated units **12**, the signal amplifier **112** may be operated with different predetermined gains, and the predetermined gains may be stored in an internal register of the signal amplifier **112**. When performing DC offset calibration for

the to-be-calibrated unit **12**, the signal amplifier **112** may read a corresponding predetermined gain from the internal register, and amplify the reference signal Sref according to this predetermined gain. In some embodiments, the maximum gain of the signal amplifier **112** may also be used as the predetermined gain. In some embodiments, the second signal Sc may have a second amplitude. As mentioned above, since the first amplitude of the first signal Sa exceeds the linear operating range of the to-be-calibrated unit **12**, clipping occurs in the second signal Sc.

(16) In some embodiments, in the to-be-calibrated unit **12**, the to-be-calibrated circuit **122** may be, but is not limited to, a digital-to-analog converter (DAC) which may be used to convert a digital signal into an analog signal. In other embodiments, the to-be-calibrated circuit **122** may also include an analog-to-digital converter (ADC), a mixer, a power amplifier (PA), a low-noise amplifier (LNA), an operation Amplifiers (OP), and a combination thereof. Further, in an embodiment where the to-be-calibrated circuit **122** is a power amplifier (PA), the power amplifier may include, for example, a variable gain amplifier (VGA) and a transconductance amplifier.

(17) In some embodiments, the second signal Sc output from the to-be-calibrated unit **12** may be an analog signal. In FIG. **1**, the second signal Sc may be sent out via the transmitter **15** and the transmitting antenna **16** to generate the first wireless signal RF**1**. Specifically, the transmitter **15** may include a mixer, a filter, and a power amplifier. The mixer may up-convert the second signal Sc to the radio frequency range, the power amplifier may amplify the up-converted signal, and the filter may filter the up-converted signal or the amplified signal for the transmitter **15** to transmit a transmitting signal Stx. The transmitting antenna **16** may transmit the transmitting signal Stx to generate the first wireless signal RF**1**.

(18) In some embodiments, the receiving antenna **17** may receive the second wireless signal RF**2** correlated to the first wireless signal RF**1**, and generate a receiving signal Srx according to the second wireless signal RF**2**, and the receiver **18** may generate the receiving signal Sr according to the receiving signal Srx. Specifically, the receiver **18** may include a low noise amplifier, a mixer and a filter. In the receiver **18**, the low noise amplifier may amplify the signal Srx, the filter may filter out the noise from the amplified signal, and the mixer may down-convert the filtered signal to generate the receiving signal Sr. In such a case, the receiving signal Sr is correlated to the second signal Sc.

(19) In some embodiments, the measurement unit **13** may include an analog-to-digital converter **130** and a harmonic measurement circuit **132**. The compensation unit **14** may include a computation circuit **140** and a memory **142**. The receiver **18**, the analog-to-digital converter **130**, the harmonic measurement circuit **132** and the computation circuit **140** may be coupled in sequence. In detail, the computation circuit **140** of the compensation unit **14** may be coupled to the DC offset calibration circuit **120** of the to-be-calibrated unit **12**, and the memory **142** may be coupled to the computation circuit **140** in the compensation unit **14**.

(20) In other embodiments, the analog-to-digital converter **130** of the measurement unit **13** may also be coupled to the to-be-calibrated circuit **122** of the to-be-calibrated unit **12** to receive the second signal Sc from the to-be-calibrated circuit **122**. In such a case, the second signal Sc is the receiving signal Sr.

(21) In a further embodiment, the analog-to-digital converter **130** of the measurement unit **13** may convert the receiving signal Sr from analog to digital form to generate a digital signal Sd, and the harmonic measurement circuit **132** may obtain magnitudes Mag of even harmonics based on the digital signal Sd. For example, the harmonic measurement circuit **132** may perform a time-domain-to-frequency-domain conversion on the digital signal Sd using fast Fourier transform or other discrete Fourier transforms to obtain the magnitudes of the fundamental frequency and other harmonics, as depicted in FIG. **4** and FIG. **6**. Description will be explained in the following paragraphs. The harmonics may include even harmonics, such as the DC component (0th harmonic), the 2nd harmonics, 4th harmonics, and others.

(22) In the above embodiment, the computation circuit **140** of the compensation unit **14** may

perform computations according to the magnitudes Mag to obtain the DC offset adjustment value OFST. Furthermore, the DC offset calibration circuit **120** of the to-be-calibrated unit **12** may change the DC level of the to-be-calibrated circuit **122** according to the DC offset adjustment value OFST, thereby changing the DC level of the to-be-calibrated unit **12**. For example, the DC offset calibration circuit **120** may include a variable resistor, which may change the resistance according to the DC offset adjustment value OFST, so as to change the DC level of the to-be-calibrated unit **12**. In some other embodiments, the DC offset calibration circuit **120** may include a DC compensation circuit, which may provide a calibration value according to the DC offset adjustment value OFST, to compensate for the DC level of the to-be-calibrated unit **12**.

(23) In the above embodiment, the DC offset calibration circuit **120** may adjust the to-be-calibrated unit **12** according to the DC offset adjustment value OFST, and consequently, the magnitude Mag of a next even harmonic measured by the measurement unit **13** may be reduced, for example, reduced to be equal to or less than a predetermined value. For example, at a time t**1**, the measurement unit **13** measures the magnitude of an even harmonic as the first magnitude, the computation circuit **140** may then obtain a DC offset adjustment value OFST as the first DC offset adjustment value. The DC offset calibration circuit **120** adjusts the to-be-calibrated unit **12** according to the first DC offset adjustment value, such that, at a subsequent time t**2**, the measurement unit **13** measure the magnitude of the even harmonic as the second magnitude and the second magnitude is less than the first magnitude. In the above embodiment, the variation trend of the magnitudes of an even harmonic may be used to assist in determining the calibration direction of the DC offset. For example, in the above adjustment process, in the case where the DC offset calibration circuit **120** is a variable resistor, the DC offset calibration circuit **120** may raise the resistance thereof according to the first DC offset adjustment value (e.g., the initial DC offset adjustment value), so that the measurement unit **13** may measure the magnitude of the even harmonic as the third magnitude at time t**3** subsequent to time t**1**. If the third magnitude is larger than the first magnitude, it may be determined that the calibration direction of the DC offset calibration circuit **120** is wrong. Consequently, the calibration direction may be changed, for example, the resistance of the variable resistor may be reduced to achieve the purpose of reducing the magnitude of the even harmonic.

(24) In the above embodiment, the memory **142** may store the predetermined value, the magnitudes Mag of the even harmonic, and the DC offset adjustment values OFST. For example, the memory **142** may store the magnitudes Mag of selected even harmonics, such as the magnitude of the 2nd harmonic, the magnitude of the 4th harmonic, the magnitude of the 6th harmonic, or the magnitudes of all the even harmonics. When the magnitude Mag of an even harmonic is larger than or equal to the predetermined value, it is determined that the to-be-calibrated unit **12** needs to be calibrated for the DC offset thereof. The computation circuit **140** may output the DC offset adjustment value OFST to the DC offset calibration circuit **120** to compensate for the DC offset. When the magnitude Mag of the even harmonic is less than the predetermined value, it is determined that the to-be-calibrated unit **12** may not need to perform a DC offset compensation, and the computation circuit **140** may not output the DC offset adjustment value OFST or may output a DC offset adjustment value OFST equal to zero to the DC offset calibration circuit **120**.

(25) In some embodiments, the predetermined value is set according to the sensitivity of the measurement unit **13**, or according to the system specification of the communication circuit where the DC offset calibration circuit **120** is used. In detail, taking the sensitivity of the measurement unit **13** for example, sine the sensitivity may be correlated to a noise level of the measurement unit **13**, the lower the noise level, the higher the sensitivity of the measurement unit **13**. The predetermined value may be set to be equal to or higher than the noise level. If the predetermined value is set to be equal to the noise level as an example, when the magnitude Mag of the even harmonic measured by the measurement unit **13** is higher than the noise level, it is determined that the even harmonic is detected, and the DC offset of the to-be-calibrated unit is required to be

calibrated. During the calibration process, the to-be-calibrated unit **12** is adjusted according to the DC offset adjustment value OFST until the magnitude Mag of the even harmonic measured by the measurement unit **13** is equal to or less than the noise level. When the magnitude of the even harmonic is less than the sensitivity (for example, less than the noise level), it is determined that the magnitude value of the even harmonic measured by the measurement unit **13** is unreliable, and it is unnecessary to calibrate the DC offset of the to-be-calibrated unit **12**.
(26) In some embodiments, the computation circuit **140** may sweep across all DC offset adjustment values OFST in a predetermined range by a predetermined step size (i.e., interval), and store the DC offset adjustment values OFST and the magnitudes Mag of a selected even harmonic in the memory **142**. For example, the predetermined step size may be 500 millivolts (mV), the predetermined range may be between −1000 mV and 1000 mV, and the computation circuit **140** may sweep five DC offset adjustment values between −1000 mV and 1000 mV at intervals of 500 mV OFST. The DC offset of the to-be-calibrated unit **12** is calibrated according to the five DC offset adjustment values OFST respectively, and the measurement unit **13** may then obtain five corresponding magnitudes Mag after each calibration. The five DC offset adjustment values OFST and the five corresponding magnitudes Mag are stored in the memory **142**. Later, the DC offset adjustment value OFST corresponding to the minimum magnitude Mag is output to the DC offset calibration circuit **120** for the DC offset compensation.
(27) The following will be explained with reference to FIGS. **3** to **6**. If there is a DC offset in the to-be-calibrated unit **12**, asymmetrical clipping will occur in the second signal Sc, i.e., the peaks and valleys of the second signal Sc are clipped to different degrees. Consequently, the receiving signal Sr will include even harmonics. On the contrary, if there is no DC offset in the to-be-calibrated unit **12**, symmetrical clipping will occur in the second signal Sc, i.e., the peaks and valleys of the second signal Sc are clipped to the same degree. Therefore, the receiving signal Sr may include merely odd harmonics without an even harmonic. The asymmetrical clipping results in even harmonics in the receiving signal Sr. Therefore, it may be determined that whether a DC offset presents in the to-be-calibrated unit **12** according to the presence of the even harmonics in the receiving signal Sr. In other words, the even harmonics in the receiving signal Sr may arise from the DC offset of the to-be-calibrated unit **12**.
(28) FIG. **3** shows a waveform of symmetrical clipping, where the horizontal axis represents the time t in microseconds (μs), and the vertical axis represents the voltage Vin volts. As shown in FIG. **3**, the to-be-calibrated unit **12** has no DC offset, and symmetrical clipping occurs in the output signal (i.e., the second signal Sc), where the peak (approximately between 1.5 us and 3.5 μs) and the valley (approximately between 6.5 us and 8.5 μs) are clipped to the same degree. FIG. **4** shows the frequency components of the symmetrically clipped waveform in FIG. **3**, where the horizontal axis represents the frequency index n and the vertical axis represents the magnitude in dB. FIG. **4** shows that the frequency components **41**, **43**, **45**, and **47** have larger magnitudes, and they represent the fundamental frequency, 3rd harmonic, 5th harmonic, 7th harmonic respectively. In this example, the magnitudes of the even harmonics are small or undetectable.
(29) FIG. **5** shows a waveform of asymmetrical clipping. As shown in FIG. **5**, there is a DC offset in the to-be-calibrated unit **12**, and the asymmetric clipping occurs in the output signal (i.e., the second signal Sc), where the peak (approximately between 1.8 us and 3.2 μs) and the valley (approximately between 6 μs and 9 μs) are clipped to different degrees. FIG. **6** shows the frequency components of the asymmetrically clipped waveform in FIG. **5**. In FIG. **6**, the frequency components **60** to **67** have larger magnitudes, and they represent the fundamental frequency, the 3rd harmonic, the 5th harmonic, and the 7th harmonic, respectively. The frequency components **60**, **62**, **64** and **66** represent the DC component, the 2nd harmonic, the 4th harmonic and the 6th harmonic, respectively. Therefore, when the to-be-calibrated unit **12** has a DC offset, the measurement unit **12** may detect even harmonics. In addition, the measurement unit **12** may detect odd harmonics regardless of symmetrical clipping or asymmetrical clipping.

(30) In the above embodiments, in order to eliminate the DC offset on the transmission path of the communication circuit **1**, there is no DC offset measurement circuit provided on the transmission path. In the communication circuit **1**, the measurement unit and the compensating unit on the receiving path may be utilized to eliminate the DC offset on the transmitting path (for example, the DC offset of the to-be-calibrated unit), resulting in reduced circuit area and manufacture cost.
(31) FIG. **2** is a flowchart of a DC offset calibration method **200** of the DC offset calibration device **10**. The method **200** includes Steps S**200** to S**208**. Steps S**200** and S**201** are used to generate a clipped signal such as the second signal Sc, and Steps S**202** to S**208** are used to calibrate the DC offset of the to-be-calibrated unit **12** according to the magnitude of an even harmonic. Any reasonable step change or adjustment is within the scope of the present disclosure. Steps S**200** to S**208** are detailed as follows: Step S**200**: The signal generation unit **11** generates a reference signal Sref and amplifies the same to output a first signal Sa; Step S**201**: The to-be-calibrated unit **12** receives the first signal Sa to output a second signal Sc, wherein the second signal Sc is clipped due to a signal saturation effect of the to-be-calibrated unit **12**, and the second signal Sc is clipped asymmetrically further due to the DC offset of the to-be-calibrated unit **12**; Step S**202**: The measurement unit **13** receives a receiving signal Sr correlated to the second signal Sc, wherein the receiving signal Sr includes even harmonics resulting from asymmetrical clipping; Step S**204**: The measurement unit **13** measures the magnitude Mag of an even harmonics; Step S**206**: The compensation unit **14** obtains the DC offset adjustment value OFST according to the magnitude Mag of the even harmonic; Step S**208**: The DC offset calibration circuit **120** adjusts the to-be-calibrated unit **12** according to the DC offset adjustment value OFST, so as to calibrate the DC offset.
(32) The explanations of Steps S**200** has been provided in the preceding paragraphs, and will not be repeated here for brevity. Steps S**201** to S**208** are explained as follows with reference to FIG. **7** and FIG. **8**. FIGS. **7** and **8** show the DC offset responses of the even harmonics at the first interval and the second interval, respectively, where the horizontal axis represents the DC offset adjustment value OFST, and the vertical axis represents the magnitude Mag.
(33) Referring to FIG. **7**, Steps S**201** and S**204** may be executed cyclically to obtain the minimum magnitude in the first predetermined range. The computation circuit **140** may sweep the first predetermined range at the first interval (i.e., step size) to obtain a coarse minimum magnitude of an even harmonic. For example, the first predetermined range may be between v**1** to v**5**, the first interval may be d**1**, and the even harmonic may be the 2nd harmonic. In the first cycle, the computation circuit **140** sets the DC offset adjustment value OFST as v**1**, and the DC offset calibration circuit **120** changes the DC level of the to-be-calibrated unit **12** accordingly. The to-be-calibrated unit **12** outputs the second signal Sc which is clipped. The measurement unit **13** receives the receiving signal Sr correlated to the second signal Sc (Step S**202**), and measures the magnitude M**1** of the second harmonic (Step S**204**). The second to fifth cycles are performed in a similar manner, so as to obtain various magnitudes of the 2nd harmonics corresponding to the various DC offset adjustment values OFST of v**2**, v**3**, v**4** and v**5**, respectively, and the various magnitudes may be M**2**, Mmin**1**, M**2** and M**1**, respectively. Thus the magnitude sweep of the 2nd harmonic for the first predetermined range is completed. In this embodiment, the memory **142** may store five DC offset adjustment values OFST (v**1**, v**2**, v**3**, v**4** and v**5**) in the first predetermined range and five corresponding magnitudes of the second harmonic (M**1**, M**2**, Mmin**1**, M**2** and M**1**).
(34) Since the magnitude Mmin**1** is less than the magnitudes M**1** and M**2**, it may be determined that the magnitude Mmin**1** may be the rough minimum magnitude in the first predetermined range, and that the preferred DC offset adjustment value OFST may be located near v**3**. According to the results of the first predetermined range, a second predetermined range is determined. The second predetermined range may be less than the first predetermined range and may cover the DC offset adjustment value OFST (i.e., v**3**) corresponding to the rough minimum magnitude Mmin**1**.
(35) Referring to FIG. **8**, the computation circuit **140** may continue to sweep the second

predetermined range at the second interval to obtain a fine minimum magnitude of the 2nd harmonic. For example, the second predetermined range may be between v**2** and v**4**, the second interval may be d**2**, and the second interval d**2** may be less than the first interval d**1**. Similarly, when the DC offset adjustment values OFST are set to v**2**, v**3-1**, v**3-2**, v**3-3** and v**4**, the corresponding magnitudes of the second harmonics of M**2**, M**3**, Mmin**2**, M**3**, and M**2** may be obtained, respectively, so as to complete the magnitude sweep of the 2nd harmonic for the second predetermined range.
(36) Since the magnitude Mmin**2** is less than the magnitudes M**2** and M**3**, it may be determined that the magnitude Mmin**2** may be the minimum magnitude in the second predetermined range, and that the preferred DC offset adjustment value OFST may be located near v**3-2**. The computation circuit **140** may compare the magnitude Mmin**2** with a predetermined value. If the magnitude Mmin**2** is equal to or less than the predetermined value, value v**3-2** may be output as the preferred DC offset adjustment value OFST to the DC offset calibration circuit **120** (Step S**206**).
(37) In the above embodiment, the magnitude sweep across the second predetermined range may be omitted. In such a case, the computation circuit **140** may compare the magnitude Mmin**1** with the predetermined value. If the magnitude Mmin**1** is equal to or less than the predetermined value, value v**3** may be output as the preferred DC offset adjustment value OFST to the DC offset calibration circuit **120**. In other embodiments, the DC offset calibration method **200** may sweep a smaller predetermined range at a smaller interval to obtain a finer minimum magnitude of the 2nd harmonic based on the principle outlined in the present invention.
(38) In Step S**208**, the DC offset calibration circuit **120** adjusts the to-be-calibrated unit **12** according to the preferred DC offset adjustment value OFST to calibrate the DC offset.
(39) In the DC offset calibration device and the DC offset calibration method according to the embodiments of the present invention, in order to eliminate the DC offset on the transmission path of the communication circuit **1**, there is no DC offset measurement circuit provided on the transmission path. In the communication circuit **1**, the measurement unit and the compensating unit on the receiving path may be utilized to eliminate the DC offset on the transmitting path, resulting in reduced circuit area and manufacturing cost.
(40) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A direct current (DC) offset calibration device comprising: a signal generation unit configured to provide a first signal; a to-be-calibrated unit coupled to the signal generation unit, the to-be-calibrated unit being configured to receive the first signal and output a second signal accordingly, wherein the second signal is a clipped signal; a measurement unit coupled to the to-be-calibrated unit, the measurement unit being configured to receive a receiving signal correlated to the second signal, the receiving signal comprising an even harmonic resulted from the clipped signal, the measurement unit being configured to measure a magnitude of the even harmonic; and a compensation unit coupled to the measurement unit and the to-be-calibrated unit, the compensation unit being configured to compute and obtain a DC offset adjustment value according to the magnitude of the even harmonic, and to adjust the to-be-calibrated unit according to the DC offset adjustment value.
2. The DC offset calibration device of claim 1, wherein the signal generation unit comprises: a signal generator configured to generate a reference signal; and a signal amplifier coupled to the signal generator, the signal amplifier being configured to receive the reference signal and amplify the reference signal to output the first signal.

3. The DC offset calibration device of claim 2, wherein: the signal generator comprises a sine wave generator and the reference signal comprises a sine wave; and the signal amplifier comprises a variable gain amplifier.
4. The DC offset calibration device of claim 1, wherein the receiving signal is the second signal.
5. The DC offset calibration device of claim 1, wherein: by a transmitting antenna, the second signal is transmitted to generate a first wireless signal; and by a receiving antenna, a second wireless signal correlated to the first wireless signal is received and the receiving signal is generated according to the second wireless signal.
6. The DC offset calibration device of claim 1, wherein the measurement unit comprises: an analog-to-digital converter configured to generate a digital signal according to the receiving signal; and a harmonic measurement circuit coupled to the analog-to-digital converter, and configured to obtain the magnitude of the even harmonic according to the digital signal.
7. The DC offset calibration device of claim 1, wherein the to-be-calibrated unit comprises: a to-be-calibrated circuit; and a DC offset calibration circuit coupled to the to-be-calibrated circuit, the DC offset calibration circuit being configured to adjust a DC level of the to-be-calibrated circuit according to the DC offset adjustment value.
8. The DC offset calibration device of claim 7, wherein: the DC offset calibration circuit comprises a variable resistor, and the DC offset calibration circuit being configured to change a resistance of the variable resistor according to the DC offset adjustment value, so as to adjust the DC level of the to-be-calibrated circuit.
9. The DC offset calibration device of claim 7, wherein: the DC offset calibration circuit comprises a DC compensation circuit configured to provide a calibration value according to the DC offset adjustment value to compensate for the DC level of the to-be-calibrated circuit.
10. The DC offset calibration device of claim 7, wherein the to-be-calibrated circuit comprises a digital-to-analog converter, an analog-to-digital converter, a mixer, a power amplifier, a low-noise amplifier, and/or an operational amplifier.
11. The DC offset calibration device of claim 10, wherein the power amplifier comprises a variable gain amplifier (VGA) and/or a transconductance amplifier.
12. The DC offset calibration device of claim 1, wherein the signal generation unit and the to-be-calibrated unit are arranged in a transmission path, and the measurement unit is arranged in a reception path.
13. The DC offset calibration device of claim 1, wherein the compensation unit is configured to adjust the to-be-calibrated unit according to the DC offset adjustment value, so as to reduce a magnitude of a next even harmonic measured by the measurement unit.
14. The DC offset calibration device of claim 13, wherein the compensation unit is configured to adjust the to-be-calibrated unit according to the DC offset adjustment value, such that the magnitude of the next even harmonic measured by the measurement unit is equal to or less than a predetermined value.
15. The DC offset calibration device of claim 14, wherein: the predetermined value is correlated to a sensitivity of the measurement unit, or to a system specification of a communication circuit where the DC offset calibration device is used; and the compensation unit comprises a memory configured to store the DC offset adjustment value and the predetermined value.
16. A DC offset calibration method, used to calibrate a DC offset of a to-be-calibrated unit, the method comprising: the to-be-calibrated unit outputting a second signal, wherein the second signal is clipped due to a signal saturation effect of the to-be-calibrated unit, and wherein the second signal is clipped asymmetrically further due to the DC offset of the to-be-calibrated unit; receiving a receiving signal correlated to the second signal, wherein the receiving signal includes an even harmonic resulting from asymmetrical clipping; measuring a magnitude of the even harmonic; computing a DC offset adjustment value according to the magnitude of the even harmonic; and adjusting the to-be-calibrated unit according to the DC offset adjustment value to calibrate the DC

offset.

17. The method of claim 16, further comprising: generating a reference signal; amplifying the reference signal to output a first signal; and by the to-be-calibrated unit, receiving the first signal and outputting the second signal accordingly.

18. The method of claim 16, further comprising: by a transmitting antenna, transmitting the second signal to generate a first wireless signal; and by a receiving antenna, receiving a second wireless signal correlated to the first wireless signal and generating the receiving signal according to the second wireless signal.

19. The method of claim 16, wherein step of adjusting the to-be-calibrated unit according to the DC offset adjustment value comprises: varying a resistance of a variable resistor according to the DC offset adjustment value to adjust a DC level of the to-be-calibrated unit; or providing a calibration value according to the DC offset adjustment value to compensate for the DC level of the to-be-calibrated unit.

20. The method of claim 16, wherein step of adjusting the to-be-calibrated unit according to the DC offset adjustment value comprises: adjusting the to-be-calibrated unit according to the DC offset adjustment value to, so as to reduce a magnitude of a next even harmonic measured by the measurement unit until the magnitude of the next even harmonic measured by the measurement unit is equal to or less than a predetermined value.